Capstone 6 – Prompt Library

Reusable prompts for the AI Coding Tutor project. Replace `<CODE>` with one of the ten programs. For each prompt, record which tool you used and what you observed.

A. Explanation prompts

ID	Prompt	Purpose	Prompt-engineering technique
PR1	"You are a patient Python tutor for a first-week beginner. Explain this code line by line in plain English, no jargon: `<CODE>`"	Line-by-line explanation	Role + audience
PR2	"For this code, make a table showing the value of every variable after each loop pass: `<CODE>`"	Trace the output	Structured output
PR3	"State the exact output this code prints, character for character. Then list any line a beginner might misread: `<CODE>`"	Output claim to verify	Constraint + self-check
PR4	"Explain `<CODE>` in 3 sentences, then in 8 sentences with an analogy."	Compare explanation depth	Length control

B. Practice-problem prompts

ID	Prompt	Purpose
PR5	"Write 3 practice problems (easy, medium, hard) based on this code. Give test cases as (input, expected output) pairs but hide the solution: `<CODE>`"	Problems that work with `check_answer()`
PR6	"Write one practice problem that tests the common mistake in this code, and say which mistake it targets: `<CODE>`"	Target misconceptions

C. Verification and safety prompts

ID	Prompt	Purpose
PR7	"Review this code for Python 2 syntax, unsafe functions (eval, exec, os.system) and bad practices. Quote each risky line: `<CODE>`"	Outdated/unsafe detection
PR8	"What could go wrong if the input is empty, zero or negative? Predict the result, don't guess: `<CODE>`"	Edge cases
PR9	"You said the output would be X. I ran it and got Y. Explain exactly where your reasoning went wrong."	Correction loop

D. Study-material prompts

ID	Prompt	Purpose
PR10	"Using only the explanations below, create a one-page beginner mini-guide with a table of 5 common traps."	Mini-guide draft
PR11	"Create a 10-question multiple-choice quiz with 4 options each, one answer key and a one-line reason for each answer."	Quiz draft
PR12	"Make a 5-slide outline: problem, method, findings, before/after, conclusion. Max 20 words per bullet."	Presentation (for Gamma)

Prompt-writing rules that worked

1. Name the audience ("first-week beginner").
2. Ask for the **exact output** separately so it can be checked.
3. Ask for a trace table; wrong claims show up as mismatched rows.
4. Never accept an answer without running the code.
5. When a claim is wrong, paste the real output back (PR9) and see whether the tool fixes it.

Usage log (fill in as you work)

Date	Program	Prompt ID	Tool	Output correct?	Notes